# Day 12 | Wednesday

November 18, 2026

12

## Percent: part of a whole

12:45 PM–2:00 PM • Math • 75 min

## Waste allowance and purchase quantities

2:15 PM–3:30 PM • Math • 75 min

**Week 4 • 2 scheduled blocks • 150 teaching minutes**

Estimate. Show the method. Label units. Check the result.

# Today’s exact schedule

## 12:45 PM–2:00 PM • 75 min

### Percent: part of a whole

| | | |
|---|---|---|
| 12:45 PM–12:50 PM | **Start** | 5 |
| 12:50 PM–12:55 PM | **Recall** | 5 |
| 12:55 PM–1:10 PM | **Teach** | 15 |
| 1:10 PM–1:25 PM | **Together** | 15 |
| 1:25 PM–1:45 PM | **Practise** | 20 |
| 1:45 PM–1:55 PM | **Check** | 10 |
| 1:55 PM–2:00 PM | **Wrap up** | 5 |

Math • Math Week 4, Lesson 5

## 2:15 PM–3:30 PM • 75 min

### Waste allowance and purchase quantities

| | | |
|---|---|---|
| 2:15 PM–2:20 PM | **Start** | 5 |
| 2:20 PM–2:25 PM | **Recall** | 5 |
| 2:25 PM–2:40 PM | **Teach** | 15 |
| 2:40 PM–2:55 PM | **Together** | 15 |
| 2:55 PM–3:15 PM | **Practise** | 20 |
| 3:15 PM–3:25 PM | **Check** | 10 |
| 3:25 PM–3:30 PM | **Wrap up** | 5 |

Math • Math Week 4, Lesson 6

**Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time**

# Percent: part of a whole

**Today’s goal: Convert percentages, decimals and fractions; find a percentage of a quantity.**

Share a prior use of this skill; identify the quantity and unit.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

Exercise Book 6A Q1–10 (p.39); 6C Q1–2 (p.43)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Recall and reconnect

Retrieve 0.5 = 1/2 and ask for the matching percent. Use Exercise Book 6A Q1 and Q3.

**Attempt the short retrieval task independently; explain the method before seeing answers.**

Resources: Exercise Book 6A Q1–10 (p.39); 6C Q1–2 (p.43)

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Recall from the Exercise Book | 6A Q1

Write 64% as a decimal.

Write the operation, working and units before checking.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Percent means parts per hundred

1. A percent compares a part with a whole using 100 equal parts.
2. Convert percent to decimal by dividing by 100.
3. Convert decimal to percent by multiplying by 100.
4. Name the whole before you calculate the part.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Find a percentage of a quantity

**1** Write the rate as a decimal.

**2** Part = decimal rate × whole quantity.

**3** The part keeps the same measurement unit as the whole.

**4** If the rate is below 100%, the part should be smaller than the whole.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Write the whole and rate before multiplying

12% of 250 mm = 0.12 × 250 = 30 mm
24 out of 300 = 24 ÷ 300 × 100 = 8%

Check the first result: 10% is 25 mm, so 12% should be slightly larger.
Do not multiply by 12 when you mean 12%.

Together: 6A Q5–8 and 6C Q2. Label part, whole and rate.
Compare decimal multiplication with the model-specific percent key. Calculate the 6C Q2 rectangle width before drawing it.

# Textbook | page 94

## Example 6.3 Brackets for a second job

A pallet holds 240 brackets. 15% of them are set aside for a second job. How many brackets is that?

**SOLUTION**

1. Change the percent to a decimal: 15% = 0.15.
2. Multiply: 0.15 × 240 = 36.

Textbook p.94 • Example 6.3 — Find a percent of a quantity

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Textbook | page 94

## CALCULATOR

### TI-30Xa

240 × 15 2nd % =

Display 36.

### Sharp EL-531XT

240 × 15 2ndF %

Display 36.

With the % key, the TI-30Xa keys end with =, and the Sharp keys end with 2ndF %. The decimal method, 0.15 × 240 =, gives 36 on both calculators too.

**Answer** **36 brackets.**

Textbook p.94 • Example 6.3 — Compare the two calculator models

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Spot the mistake. Explain the correction.

## A learner enters 25 × the whole to find 25% of it. What is missing?

Discuss the method before checking the answer.

**The percent must become a decimal: 25 ÷ 100. Multiply the whole by 0.25.**

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Work together

Work Exercise Book 6A Q5–8 and 6C Q2. Label the part, whole and rate.

**Work in pairs; alternate calculator operator and checker. Show units and an estimated answer.**

Resources: Exercise Book 6A Q1–10 (p.39); 6C Q1–2 (p.43)

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Guided practice | 6A Q5

Write 45% as a fraction in lowest terms.

Work with a partner. Take turns explaining the next step.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Compare your method | 6A Q5

Write 45% as a fraction in lowest terms.

**CHECK TOGETHER**

**9/20**

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Your scheduled practice

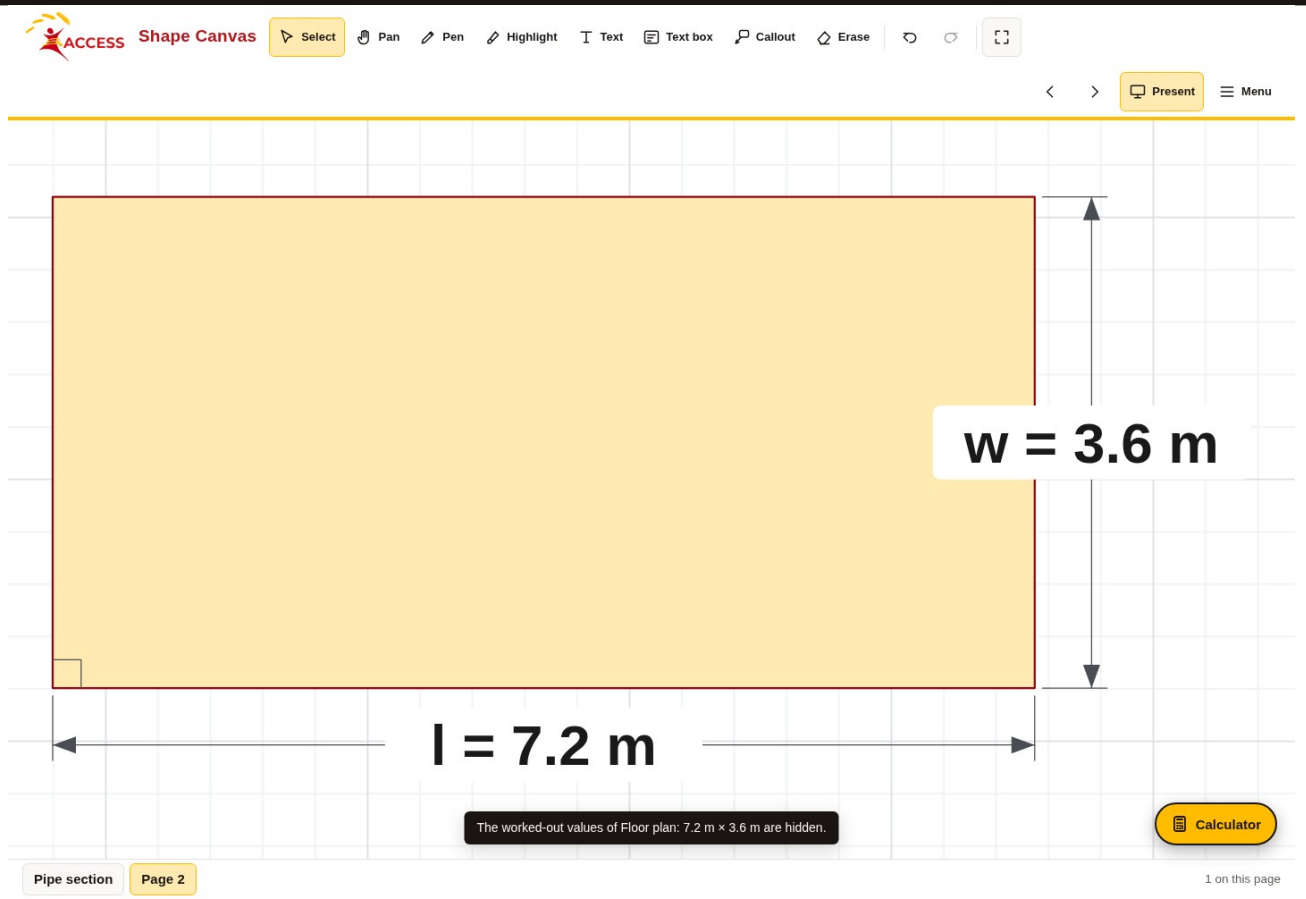
Complete Exercise Book 6A Q2,4,9–10 and 6C Q1. Create the 6C Q2 rectangle in Shape Canvas after calculating its missing width.

**Solve on paper first; use the tool to check; correct one error with a written reason.**

Resources: Exercise Book 6A Q1–10 (p.39); 6C Q1–2 (p.43)

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Open Shape Canvas




Navigation example. Use the values and units in the assigned task.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Practise in Numeracy Hub and Shape Canvas

1. Label the part, whole and rate in the assigned book question before using a tool.

2. Calculate the missing width for 6C Q2, then create its rectangle in Shape Canvas.

3. Check with the matching Numeracy Hub calculator model and compare the methods.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Practise from the Exercise Book | 6A Q2

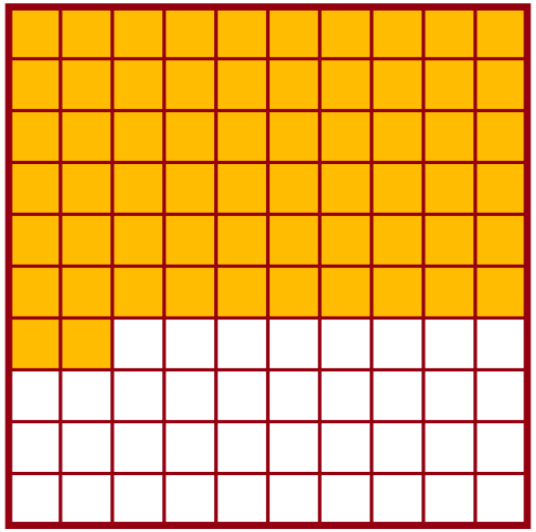
Write 3.5% as a decimal.

Write the operation, working and units before checking.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Practise from the Exercise Book | 6C Q1

Write the shaded part as a percent, a decimal and a fraction in lowest terms.



Write the operation, working and units before checking.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Check independently

Exercise Book 6A Q8–9 with a labelled part/whole calculation.

**Solve without displayed answers; include working, units and the requested rounding.**

Resources: Exercise Book 6A Q1–10 (p.39); 6C Q1–2 (p.43)

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Independent check | 6A Q8

Find 6.5% of 4,200.

Answer independently. Keep worked solutions closed.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Independent check | 6A Q9

27 is what percent of 360?

Answer independently. Keep worked solutions closed.

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Wrap up and choose the next step

Ask: What did you learn? Where will you use it? What will you check next time?

**Name the math and one other essential skill used; write the next practice step.**

Resources: Exercise Book 6A Q1–10 (p.39); 6C Q1–2 (p.43)

Textbook sections 6.1–6.3 pp.92–95; calculator percent p.35

# Break

2:00 PM–2:15 PM

Return ready for: Waste allowance and purchase quantities

**This break is excluded from teaching minutes.**

# Waste allowance and purchase quantities

**Today’s goal: Apply an allowance or discount once, then round an order to whole purchase units.**

Share a prior use of this skill; identify the quantity and unit.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

Exercise Book 6A Q11–13 (pp.39–40); 6B Q1–3,5 (p.41)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Recall and reconnect

Discuss whether a 15% allowance means ordering 15% or 115% of the original quantity.

**Attempt the short retrieval task independently; explain the method before seeing answers.**

Resources: Exercise Book 6A Q11–13 (pp.39–40); 6B Q1–3,5 (p.41)

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Apply an allowance to the original amount

1 First identify the net quantity before any allowance.

2 An increase of r% gives net quantity × (1 + r/100).

3 A discount of r% gives original amount × (1 − r/100).

4 Use the percentage stated in the exercise; it is problem data.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Turn a calculation into an order

1. Keep the unrounded quantity after applying the allowance.
2. Divide by the quantity supplied in one purchase unit if needed.
3. Round up once at the final step to buy enough whole units.
4. Show net quantity, allowance and final order separately.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Allowance first; whole-unit order last

6B Q2: 18 sheets × 1.15 = 20.7 sheets
Whole sheets to purchase = 21

**Check another way: allowance = 18 × 0.15 = 2.7 sheets;**
**18 + 2.7 = 20.7. Do not add another 15% to this result.**

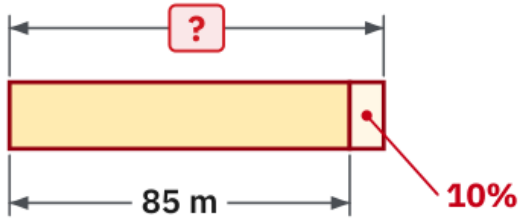
Together: 6A Q11–13 and 6B Q3.
Identify the original base before calculating a discount. Compare percent-key and decimal-multiplier methods.

All allowances here are problem data—not standard job allowances.

# Textbook | page 96

## Example 6.5 Ordering pipe with a waste allowance

A job needs 85 m of pipe. Add a 10% waste allowance. The pipe is sold in 6 m lengths. How many lengths do you order?




**SOLUTION**

1. 10% of 85 = 0.1 × 85 = 8.5 m.
2. Add it on: 85 + 8.5 = 93.5 m.
3. Lengths: 93.5 ÷ 6 = 15.5833...
4. A part length still has to be bought, so round up to 16.

Textbook p.96 • Example 6.5 — Allowance and whole purchase lengths

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Textbook | page 96

**CALCULATOR**

**TI-30Xa**

85 | + | 10 | 2nd | % | =

Display 93.5

**Sharp EL-531XT**

85 | + | 10 | 2ndF | %

Display 93.5

Pressing + and then % adds 10% of 85 to 85 on both calculators (TI-30Xa guidebook, p. 4, and Sharp EL-531XT Calculation Examples [4]). In one step without the % key: 85 × 1.1 = 93.5.

**Answer** **93.5 m of pipe, which is 16 lengths.**

Textbook p.96 • Example 6.5 — Calculator check and final order

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Spot the mistake. Explain the correction.

## Does a 15% allowance mean ordering 15% of the original quantity?

Discuss the method before checking the answer.

**No. The original 100% plus 15% gives 115% of the net quantity. Apply the allowance once.**

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Work together

Work Exercise Book 6A Q11–13 and 6B Q3; learners identify the original base before calculating.

**Work in pairs; alternate calculator operator and checker. Show units and an estimated answer.**

Resources: Exercise Book 6A Q11–13 (pp.39–40); 6B Q1–3,5 (p.41)

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Guided practice | 6A Q11

Increase 340 by 15%.

Work with a partner. Take turns explaining the next step.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Compare your method | 6A Q11

Increase 340 by 15%.

**CHECK TOGETHER**

**391**

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Your scheduled practice

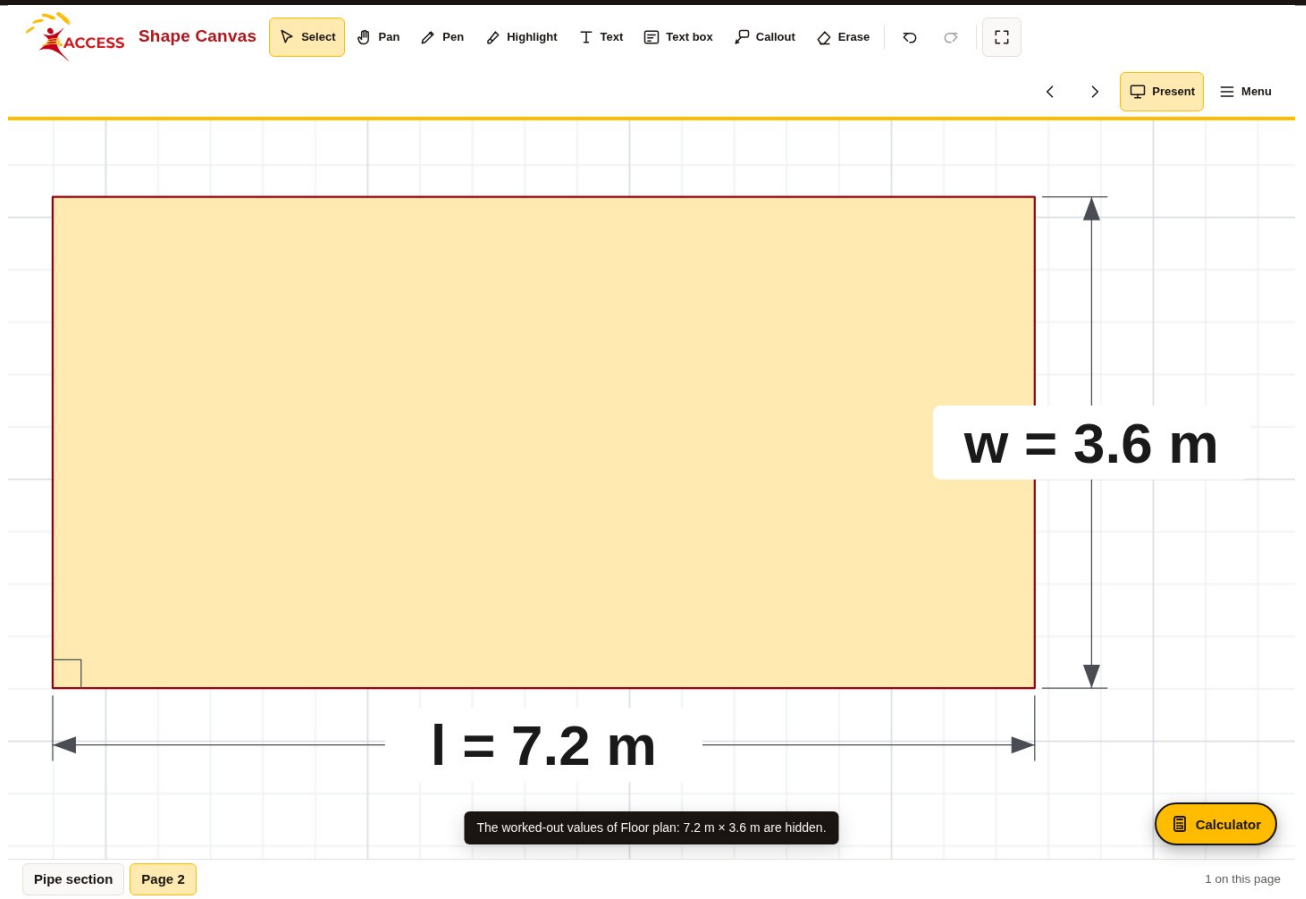
Solve Exercise Book 6B Q1,5 and rework Q2 with a written order explanation. Label all practice allowances as problem data, not a standard job allowance.

**Solve on paper first; use the tool to check; correct one error with a written reason.**

Resources: Exercise Book 6A Q11–13 (pp.39–40); 6B Q1–3,5 (p.41)

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Open Shape Canvas




Navigation example. Use the values and units in the assigned task.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Practise in Numeracy Hub and Shape Canvas

1. Label a Shape Canvas rectangle with the net amount and its allowance from the exercise.

2. Compare the percent-key and decimal-multiplier methods in Numeracy Hub Calculator.

3. Record the unrounded amount, then the whole-unit order with your rounding reason.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Practise from the Exercise Book | 6B Q1

A pipe run needs 64 m of pipe. Add a 12% waste allowance. The pipe on this job comes in 6.4 m lengths. How many metres do you need, and how many lengths do you order?

Write the operation, working and units before checking.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Practise from the Exercise Book | 6B Q5

Of 85 welds tested, 6 are rejected. What percent are rejected, to two decimal places?

Write the operation, working and units before checking.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Check independently

Exercise Book 6B Q2 and Q3, showing the unrounded order and discount calculation.

**Solve without displayed answers; include working, units and the requested rounding.**

Resources: Exercise Book 6A Q11–13 (pp.39–40); 6B Q1–3,5 (p.41)

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Independent check | 6B Q2

A fabrication job needs 18 sheets of 4 ft × 8 ft plate. Add a 15% allowance and round up to whole sheets.

Answer independently. Keep worked solutions closed.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Independent check | 6B Q3

A bearing lists at $86.50. Your shop gets a 12% trade discount. What does one bearing cost? What do 6 cost? Round to the cent.

Answer independently. Keep worked solutions closed.

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87

# Wrap up and choose the next step

Ask: What did you learn? Where will you use it? What will you check next time?

**Name the math and one other essential skill used; write the next practice step.**

Resources: Exercise Book 6A Q11–13 (pp.39–40); 6B Q1–3,5 (p.41)

Textbook sections 6.4 pp.96–97; review sections 5.7 pp.86–87